

→ Role of computer vision in Robotics

Computer vision gives robot the availability to see, understand & act. By processing image from camera, robot can recognise object, locate themselves, & plan action such as picking, avoiding or navigating.

→ Key Function of Computer vision

→ Object Recognition

Robot identify & classify object (cup, bottle, tool) using visual feature or learned pattern

→ Localisation → ~~By analysing visual~~

By analyzing visual cues, robots determine their own location & orientation within space (visual SLAM)

Basic camera projection Formula

$$S \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R]T \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

where,

(x, y, z) = 3D world point

(u, v) = 2D image point

K = camera intrinsic matrix

R, t = rotation & translation (robot pose)

*

It shows how robot convert real 3D points → Camera Image co-ordinates, enabling

3. Trajectory Planning —

Robots calculate safe & efficient path to reach or avoid object, adapting to dynamic environments

Conventional Recognition Method vs Deep learning Based Methods.

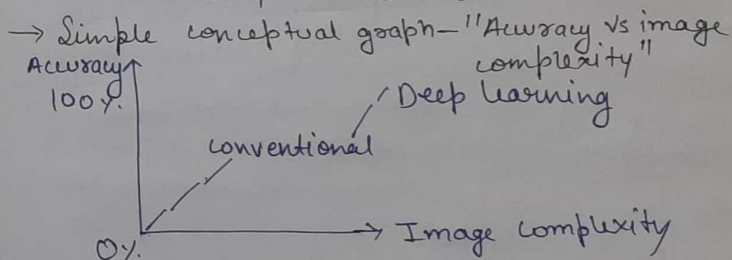
Aspect	Conventional Methods	Deep learning Method
Approach	Hand crafted feature + Simple Classifier	learns feature automatically using CNN's
Accuracy	Good only in control condition	High accuracy in complex, real setting
Efficiency	Fast but limited for complex task	need high compute but handle complexity well
Flexibility	Hard to adapt, feature must be redesigned	Easily adaptable with retraining
Limitation	weak generalization, struggle with noise	Requires big dataset, less interpretable

Example in real Robot -

- Self driving cars - detect road, people, sign
- Industrial Robot - pick & place, using camera feedback.
- Home robots - Recognise household objects & navigate safely.

How visual perception Aids Robots -

- Decision-Making → Robot use to vision to decide their next action, such as grasping an object, avoiding obstacle or following a person.
- Engagement - visual perception allow robot to interact naturally with their surrounding, adapting to changes & unexpected event
- Learning - Robot can improve their performance over time by learning from visual feedback, making them more autonomous & versatile.



This show that conventional method drop in accuracy as scene become complex (clutter, occlusion), while deep learning stay more stable.

→ Roles of Sensors & actuators in an autonomous warehouse robot →
Sensors & actuators enable a warehouse robot to perceive its environment & perform action. Such as navigating, lifting or transporting item.

→ Contribution of Sensor

- Obstacle Detection: Sensors like LIDAR, ultrasonic, IR & depth camera allow the robot to detect shelves, walls & work to avoid collision.
- Object Handling - vision sensor help identify objects, read barcode/labels, estimate size, shape, and orientation for precise picking.
- Navigation - IMU, wheel encoders & LIDAR support map building, localization & path following in dynamic warehouse layouts.

→ Contribution of Actuators →

- Mobility - Motors & wheel leg convert control signal into movement, enabling smooth navigation & turning.
- Manipulation → Servo Motor & grippers allow robot to pick up, rotate & place object reliably.

Precision Action → Actuator involve controlled motion during stacking, sorting or alignment tasks.

Benefits of Sensor →

- * High accuracy environment understanding
- * Enables safe navigation around workers & obstacles.
- * Support object identification & tracking
- * Real time mapping (SLAM) for dynamic environment

Draw back of Sensor

- * Sensitive to lighting, dust, & reflective surface (camera, LIDAR)
- * High computational load for sensor fusion
- * possible signal noise causing mis detection.

2 Benefits of actuators →

- * High precision movement & repeatability
- * Smooth control of speed & torque
- * Reliable for Repetitive warehouse task.

Drawback of actuators

- * Energy consumption increasing with payload
- * wear & tear requiring maintenance
- * precision reduction under heavy loads

Difficulties in combining Sensors & Actuators

- Sensor fusion → Synchronising different sensors & removing noise is difficult.
- Precision → small error affect picking accuracy & docking.
- Reaction time — Fast warehouse require quick processing & actuator response

Motion planning in Robotic Navigation

Motion planning decide a collision free feasible path for a robot to move from start to goal while considering obstacle, robot limit & the environment.

→ Function of Motion planning

- * Generate Safe & Smooth path
- * Avoid obstacle & ensure efficient navigation
- * Consider robot kinematic, dynamic & workplace layout.

→ Difference b/w planning technique

- * Graph based (Dijkstra A*) - optimal, good for static map but Slow in dynamic areas.
- * Sampling based (PRM-RRT) - Suitable for complex space; Fast but may give non optimal path.
- * Optimisation based → Smooth dynamic feasible trajectory (e.g. MPC).
- * Reactive Method → Fast local avoidance but can get stuck in ^{traps}

Effectiveness & Adaptability-

- * Graph method = accurate
- * Sampling = flexible for large environment
- * Optimisation = smooth real robot motion
- * Reactive = quick adoption to sudden obstacle.

→ Difficulties in Motion planning

- * Dynamic obstacle change unpredictably
- * Non holonomic constraints limit motion direction
- * High consumption for complex space
- * Real time update required for safety

→ Impact of AI

- * Deep learning improve perception & decision making
- * Reinforcement learning help robot learn navigation policies
- * Hybrid AI + classic planner give faster, safe motion in dynamic environment.

Conclusion →

Motion planning is essential for robot navigation. Each technique has benefits. Each technique has benefits & limitation & AI is making planning more accurate, adaptive real time.